Version Control System (VCS)

Q. What is mean by Version Control System?

-> Any software or any system that manages source code changes over time is nothing but Version Control System.

-> Version control is also called source control.

-> Purpose of Version Control System is to maintain versions of Project Files.

-> Examples of Version Control tools : Git,SVN,AccuRev,Aws CodeCommit,Azure DevOps Server

Q. Why Version Control System?What is Need of Version Control System? What are Advantages of Version Control System?

-> Key Reasons for using Version Control System.

1. We can work in collaboration.

-> Multiple Sotware Engineers can work on same project at a time without overwritting each other changes.

2. We can maintain source code history.

-> Every change made in the source code is recorded including who,when and why?

3. We can Overcome Conflicts

-> One developer can't overwrite other developer code changes.Bymistake if you are overwritting other developer changes then you will get conflict. If there is conflict then you can't merge your changes in branch.First,resolve the conflict and then merge your code changes in the branch.

4. We can Restore code changes to Previous Version if there is problem.

-> If added changes introduces a bug or an error then we can revert the changes and restore the source code to previous version.

5. We can perform developement in insolated manner.

-> VCS allows developer to experiment with new ideas in seperate branches without affecting the main codebase.

6. We can perform feature Developement in parallel.

-> CVS allows developer to work on new features in parallel and later that new features can be integrated with main codebase.

7. We can perform code review.

-> CVS allows us to perform code review through the pull request. i.e Developed code get reviewed by another developer through the pull request.

8. Support modern CI(Continuous Integration)/CD(Continuous Deployment) pipelines

-> CVS system support modern CI/CD pipelines

Types of Version Control System

-> There are mainly two types of version control Systems

1. Centralized Version Control System(CVCS)
2. Distributed Version Control System(DVCS)

Q. What is Centralized Version Control System (CVCS)?

-> In Centralized version control system all versions of files are stored at single place.

-> Centralized Version Control System have client-server architecture.

-> Server acts as main repository or Central Repository.

-> Client can access the source code,do some modification and again push the changes on server.

-> The main advantage of Centralized version control system is easy to manage and easy to use but main disadvantage of Centralized version control system is single point of failure. If server is down then total work get stopped.

Examples: SVN (Subversion), CVS(Concurrent Version Control System)

Q. What is distributed version Control System?

-> In Distributed version control system each client has local copy of central repository having same source code as server.

-> Examples: Most widely used distributed Control system Tool is Git.

1) Google uses "Piper" version control System.
2) AWS uses "Aws CodeCommit" Version Control System.
3) Alternative to git is "Azure DevOps Server"

Q. What is difference between Centralized Version Control System(CVS) and Distributed Version Control System.?

-> 1. In CVS every time client need to get local copy of source code from server but in DVS no need to get local copy of source code every time because every client has local copy on his system.

2. In CVS client modify the source code and commit the changes directly on central repository but in DVS client modify the source code and commit the changes in local repository first and then push the changes to central repository.

3. CVS System is easy to learn and easy to set up but DVS system is difficult for beginners because multiple commands need to remember.

4. Working on branches is difficult in CVS because developer often faces merge conflict but working on branches is easy in DVS because developer faces lesser conflict.

5. CVS system does not provide offline access but DVS system provide offline access to local repository

6. CVS system is slower because every command hit server but DVS is faster because user is communicating with local repository.

7. If CVS server is down then developer can't work but in DVS If server is down then developer work using local repository.

Q. How to deal with your daily notes?

-> First you need to clone "DAILY-NOTES" Centrel repository on local system by using command "git clone repository_ul"
git clone https://github.com/RRDTNP/DAILY-NOTES.git
->This is one time activity.
-> Next time onword, you need to use "git pull" command.

-> How to use "git pull" command?
-> Open your local repository "DAILY-NOTES" i.e open "DAILY-NOTES" folder.
-> Right click and select "open git bash here"
-> in git bash type a command "git pull" and press enter.
-> Important Note:
If we want to read text file,first copy that text file and paste somewhere location and from that location open that file.